

UIU DOCK

Resource Center, Virtual Study Rooms & Help Desk

Official Project Proposal

A database-driven course resource center, online group study and course-wise help platform

Course

Database Management System Lab

Submitted To

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Introduction

In university education, effective learning depends not only on lectures but also on access to organized resources and consistent peer collaboration. Students need a reliable place to find verified course materials — lecture notes, slides, question banks, lab sheets, and reference links — and a structured way to study together and discuss course-specific problems. Without such infrastructure, learning becomes fragmented and heavily dependent on informal, scattered sharing.

Currently, course resources are often dispersed across personal drives, senior batches, scattered messaging groups, and unorganized social media posts. There is no single searchable hub where resources are categorized by course, subject, semester, and type, with version control, ratings, and authenticity. Similarly, online group study is usually limited to ad-hoc video calls without course-specific rooms, scheduling, attendance, or shared materials. For day-to-day problem solving, students lack a dedicated course-wise discussion space — general groups mix many courses together, so course-specific questions and solutions get lost.

UIU Dock is proposed as a centralized, database-driven web platform that unifies three core academic needs into one system: **(1) Resource Center / Hub** for course-wise resource sharing and discovery, **(2) Online Group Study Sessions / Rooms** for course-specific collaborative learning, and **(3) Help Desk Center** implemented as **course-wise group chats** where students select a course and join its dedicated chat to discuss problems and solutions. The platform will provide role-based access for students and administrators. All entities — students, courses, resources, study rooms, sessions, participants, and chat messages (room chats and course help chats) — will be stored in a normalized relational database with proper keys, constraints, and relationships.

From a Database Management System perspective, the project is designed to demonstrate strong relational modeling, normalization, full-text search, real-time data handling, and analytical reporting. It will showcase complex queries such as most-accessed resources per course, active study rooms, most active course help chats, participation trends, and contributor leaderboards, while maintaining data integrity, security, and auditability suitable for a Database Management System Lab project.

Objectives

- To provide a centralized Resource Center / Hub where course-wise study materials (notes, slides, past questions, lab resources) can be uploaded, categorized, searched, and validated.
- To enable students to create, discover, and join course-specific Online Group Study Rooms / Sessions with scheduling, capacity control, and shared resource access.
- To establish a Help Desk Center as course-wise group chats where students select a course, enter that course's dedicated chat, and collaboratively discuss problems, share solutions, and help each other with real-time messaging and persistent history.
- To organize academic collaboration by linking courses, subjects, resources, study rooms, and course help chats through a normalized relational structure.
- To support real-time communication and collaboration using database-backed chat, room presence, membership history, and notifications.
- To ensure content quality through moderation tools, ratings, reviews, reporting, versioning, and role-based access control for resources and chat conduct.
- To provide administrators with dashboards for managing users, courses, resources, rooms, and course chat activity, along with analytics and audit trails.

Project Type

Web Application - A database-driven Course Resource Center, Virtual Study Room and Course-wise Help Desk Platform.

UIU Dock will be developed as a full-stack web application accessible through a browser. The platform will be centered around three integrated modules: a searchable Resource Hub for course materials, course-specific Online Study Rooms for scheduled group sessions, and a Help Desk Center structured as **course-wise group chats** —

students browse the list of courses, select a course, and enter that course's dedicated group chat to exchange questions, explanations, and solutions. Students will browse and contribute resources, join or host study rooms, and participate in course chats; administrators will govern users, courses, content, and system analytics.

Problem Statement

Students often struggle to locate authentic and up-to-date course materials. Resources are scattered across different batches, drives, and messaging groups with no standard categorization by course code, subject, semester, or resource type. Important files get lost in chat histories, duplicate or outdated versions circulate, and there is no rating or verification mechanism to identify trustworthy materials. This wastes time and affects preparation quality.

For collaborative learning, there is no dedicated academic space to host course-specific online study sessions. Generic meeting links lack course context, scheduling, room discovery by subject, attendance tracking, shared resource panels, or post-session summaries. As a result, finding or organizing a relevant study session — especially with unfamiliar classmates from the same course — remains difficult.

For quick academic help, students need course-focused discussion but typically only have large, mixed-course messaging groups. Questions about a specific course are buried under unrelated conversations, there is no persistent course-wise chat history, no way to list all available courses and jump into the relevant course's group chat, and no searchable archive of past problems and solutions for that course. Repeated questions and lack of organized peer discussion reduce learning efficiency.

UIU Dock addresses these gaps by proposing a single database-driven platform that consolidates verified course resources in one searchable hub, provides discoverable course-wise virtual study rooms with scheduling and participation management, and offers a Help Desk Center where each course has its own persistent group chat — students simply select a course from the course list and enter that course's group chat to talk about problems and solutions in real time. The goal is to make course learning more organized, collaborative, and accessible.

Deliverables

The expected deliverables of UIU Dock include the following:

- **Resource Center / Hub:** Central repository for course-wise resources (notes, slides, past questions, lab sheets, links) with categories, tags, versioning, file/link support, search & filtering by course / subject / semester / type, ratings, and download tracking.
- **Resource Upload & Moderation:** Students can upload resources; moderators/admins verify, approve/reject, manage versions, and remove reported or outdated content with audit logs.
- **Course & Subject Management:** Structured catalog of courses and subjects with semester mapping, enabling accurate resource, room, and help-chat organization and discovery. Course list is the entry point to the Help Desk.
- **Online Group Study Rooms / Sessions:** Course-specific virtual study rooms with creation, scheduling (date/time/duration), capacity, description, resource attachments, join/leave, host controls, presence/attendance, and session history.
- **Room Discovery & Participation:** Search and filter rooms by course, subject, schedule, availability, and host; manage participants, waiting lists, and session reminders.
- **Help Desk Center — Course-wise Group Chats:** A Help Desk organized as persistent group chats per course. Students view the list of courses, select a course, and enter that course's group chat to post questions, share solutions, discuss problems, and collaborate. Features include real-time messaging, typing indicators, online presence, join/leave history, pinned messages, and searchable chat archives per course.
- **Real-Time Chat & Notifications:** Database-persisted messaging for study rooms and course help chats (Socket-based), with in-app and/or email notifications for room invites, new messages in joined course chats, resource approvals, and upcoming sessions.
- **User Authentication & Role-Based Access:** Secure registration/login with JWT/bcrypt and roles for Student and Administrator with appropriate permissions (course chat membership is tracked per user per course).

- **Moderation & Reporting Dashboard:** Tools to report inappropriate resources, rooms, or chat messages; moderators can review reports, remove content, manage course chat conduct, and maintain quality.
- **Admin Dashboard & Analytics:** Manage users, courses, resources, rooms, and course chats; view analytics such as most-downloaded resources per course, most active study rooms, most active course help chats (message volume), peak discussion hours, top contributors, and participation trends.
- **Database Design & Documentation:** Normalized relational database with ERD, primary/foreign keys, constraints, indexes for search, full-text support, and complex query demonstrations across resources, rooms, and course chats.

Project Overview

UIU Dock is a web-based academic collaboration platform that integrates a resource repository, virtual study rooms, and course-wise help chats into one database-driven system. Students register, create profiles with enrolled courses, and interact through a unified dashboard where all activities are course-contextualized. All data — users, courses, resources, rooms, participants, and chat messages (room chats and course help chats) — is stored centrally to enable searchability, linking, and analytics.

For the Resource Center, students can browse resources by course or subject, filter by type/semester, preview details, download or bookmark, rate and review, and track version history. Contributors can upload files or links with metadata (title, course, subject, type, description, tags). Moderators verify submissions before they become publicly searchable, ensuring quality and authenticity. Popular and recently added resources are highlighted via database queries.

For Online Group Study Rooms, any student can host a course-specific room by defining course/subject, topic, schedule, duration, maximum participants, study mode, and attached resources. Other students can discover rooms via course filters, view details, join/leave, and receive reminders. Each room has a persistent chat, participant list, shared resource panel, and attendance history. Hosts can manage participants, update room details, and close or archive sessions after completion.

For the Help Desk Center, the system provides a browsable list of all courses. Selecting a course takes the student into that course's dedicated group chat — a persistent, real-time chat room for that course where all enrolled/interested students can discuss problems and solutions together. Students can post questions, reply in threads, share code snippets or resource links, pin important solutions, and search past conversations. Chat history is persisted per course, membership and message metadata are tracked, and users receive notifications for new messages in their joined course chats. This creates a searchable, course-specific knowledge archive built organically by peer discussion.

For administrators, the system provides comprehensive management of users, courses, resources, rooms, and course help chats, including approvals, reports, moderation of messages, user management, announcements, and platform-wide analytics. The platform supports both synchronous (live chat/rooms/course chats) and asynchronous (resource access/chat history search) learning, with all interactions audited in the relational database.

Benchmark Analysis

UIU Dock vs Existing Platforms

Feature	UIU Dock (Our Project)	UIU LMS (eLMS)	BRACU Vault	Onebyzero Edu	StudyDock	Academizz	Notaloy
Resource Center / Hub Notes, slides, books, past questions, labs, links with categories, tags, versioning, search, ratings, download tracking	✓ Yes (All features)	✓ Yes (Course materials only)	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes
Resource Upload & Moderation Upload resources, verify, approve/reject, version control, audit logs	✓ Yes (With versioning & audit logs)	✗ No (Student upload not allowed)	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes
Course & Subject Management Catalog of courses & subjects with semester mapping	✓ Yes (Semester mapped)	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes
Online Group Study Rooms / Sessions Create, schedule, attach resources, capacity, host controls, presence, join/leave, session history	✓ Yes (All features)	✗ No (No)	✗ No (No)	✗ No (No)	✗ No (No)	✗ No (No)	✗ No (No)
Room Discovery & Participation Search/filter rooms, view schedule & availability, host rooms, manage participants, waiting lists, reminders	✓ Yes (All features)	✗ No (No)	✗ No (No)	✗ No (No)	✗ No (No)	✗ No (No)	✗ No (No)
Help Desk Center — Course-wise Group Chats Persistent group chats per course for questions, discussion & support	✓ Yes (Persistent per course)	✗ No (No)	✗ No (No)	✗ No (No)	✗ No (No)	✗ No (No)	✗ No (No)
Real-Time Chat & Notifications Real-time messaging, typing indicators, online presence, join-leave history, pinned messages, searchable chat archive	✓ Yes (Socket-based)	✗ No (No)	✗ No (No)	✗ No (No)	✗ No (No)	✗ No (No)	✗ No (No)
User Authentication & Role-Based Access Secure login (JWT/bcrypt), Student & Administrator roles, course-wise access control	✓ Yes (Roles & permissions)	✓ Yes (University login)	✗ No (Open access)	✗ No (Open access)	✗ No (Open access)	✓ Yes (Registration)	✗ No (Open access)
Moderation & Reporting Dashboard Report inappropriate content (resources, rooms, chats), review, take action, maintain quality	✓ Yes (Reports & actions)	✗ No (No)	✗ No (Manual / Limited)	✓ Yes (Limited)	✓ Yes (Limited)	✓ Yes (Limited)	✓ Yes (Limited)
Admin Dashboard & Analytics Analytics on resources, rooms, chats, users, engagement, contributions, trends & activity insights	✓ Yes (Advanced analytics)	✓ Yes (But limited for students)	✗ No (N/A)	✓ Yes (Basic)	✓ Yes (Basic)	✓ Yes (Basic)	✗ No (N/A)
Database Design & Documentation Normalized database (ERD), indexes, constraints, full-text search, complex queries support	✓ Yes (ERD, indexes, full-text search)	✗ No (No)	✗ No (No)	✗ No (No)	✗ No (No)	✗ No (No)	✗ No (No)
Mobile Friendly / Responsive Optimized for all devices and screen sizes	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes
Ratings & Reviews Rate & review resources, rooms & platform experience	✓ Yes (Resources, rooms)	✓ Yes (Course materials)	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes
Announcements & Updates Platform-wide and course-wise announcements & updates	✓ Yes (Platform & Course-wise)	✓ Yes (Course-wise)	✗ No (No)	✓ Yes (Limited)	✓ Yes (Limited)	✓ Yes (Limited)	✗ No (No)

★ UIU Dock is the only platform that combines Resource Center, Online Group Study, and Help Desk (Group Chat) with complete features in one place for UIU students.

Scope

The scope of UIU Dock covers the design and development of a centralized web platform integrating three pillars: Resource Center/Hub, Online Group Study Rooms, and Help Desk Center as course-wise group chats. The system will support user registration/login, role-based access, profile management, course/subject catalog, resource upload/search/download/rating/versioning, study room creation/discovery/scheduling/participation/chat, course list browsing and per-course group chat participation with persistent history, notifications, moderation, and analytics.

The database will follow a normalized relational model with strong entity relationships and constraints linking users ↔ courses ↔ resources ↔ rooms ↔ participants ↔ messages ↔ course help chats. It will include indexes for course-wise search and full-text search on resource titles/descriptions and course chat message archives. The project will deliver a functional academic prototype demonstrating end-to-end workflows: resource contribution and discovery, room scheduling and participation, and course help chat discussion with searchable history, plus database-driven reporting.

Out of scope for the academic timeframe are full Learning Management System (LMS) features such as automated grading, AI tutoring bots, integrated video-conferencing infrastructure, payment gateways, plagiarism detection, or direct integration with university student information systems. Live video/audio will be handled via external meeting links attached to rooms rather than a built-in SFU; advanced AI recommendations and native mobile apps will also remain future work. Private one-to-one tutoring and senior-assigned queues are also out of scope — the Help Desk is intentionally many-to-many per course, not a request-assignment system. The focus remains on a robust web prototype with a strong relational database foundation and real-time chat persistence per course.

Technology Stack

UIU Dock will be developed as a full-stack web application using modern web technologies. The frontend will offer responsive interfaces for resource browsing, room management, and course-wise help chats. The backend will expose REST APIs for auth, courses, resources, rooms, and course chats, plus WebSocket services for real-time messaging, presence, and notifications.

The database will use a normalized MySQL relational model with primary/foreign keys, constraints, transactions, and support for full-text search and complex analytics. File resources will be stored via controlled storage (local or cloud) with metadata and access logs in the database. Development and testing will use version control and API/database/realtime testing tools.

Technology Stack — Proposed Components

Area	Technology
Frontend	HTML, CSS, JavaScript / React (Responsive UI for Hub, Rooms, Course Help Chats)
Backend	Node.js, Express.js (REST APIs + WebSocket server)
Real-time Communication	Socket.IO / WebSockets (room chat, course help chat, presence, typing, notifications)
Database	MySQL (Normalized, Indexed, Full-Text Search on resources & chat)
File / Resource Storage	Local storage / Cloud storage (e.g., Cloudinary / S3) with DB metadata
Database Design Tools	MySQL Workbench / DBeaver
Authentication	JWT, bcrypt, Role-Based Access Control
API & Realtime Testing	Postman / Insomnia, Socket.IO client tools
Version Control	Git, GitHub
Local Development	XAMPP or Node.js dev environment
Notifications	In-app (DB-persisted) + Email/SMTP for offline alerts

Risks and Dependencies

- **Scope Management Risk:** Three pillars (Resource Hub, Study Rooms, Course-wise Help Chats) increase scope. Core workflows — resource upload/search, room create/join/schedule, course selection → join group chat and discuss — will be prioritized; advanced features like thread summaries or built-in video will be deferred.
- **Database Complexity Risk:** Relationships among users, courses, resources, rooms, participants, and course chat messages are extensive (notably many-to-many membership in course chats). Mitigated via careful ERD design, normalization (3NF), surrogate keys, foreign keys, constraints, indexes, and transaction handling for concurrent chat/room joins.
- **File Storage & Content Quality Risk:** Unverified or duplicate resources and noisy course chats may degrade trust. Mitigated with resource moderation workflow (pending → approved/rejected), versioning, reporting, ratings, duplicate detection via hashing, chat moderation (reporting, pinning, search), and storage quotas; DB stores metadata/audit while files reside in managed storage.
- **Real-Time Service Dependency:** Course help chats and room chats depend on WebSocket stability. All messages are persisted in the database so history is never lost; reconnection logic and fallback polling will be implemented; external meeting links for rooms remain optional.
- **Limited Real-World Data Access:** Prototype may not connect to university SIS. Realistic sample courses, resources, rooms, and course chat histories will be seeded for testing and query demonstration.
- **Security & Privacy Risk:** Resources and chats contain sensitive academic data. Mitigated with JWT auth, password hashing, RBAC, input validation, rate limiting, XSS/SQL-injection prevention, chat rate limiting, and appropriate DB access controls and audit logs.
- **Technical Integration Risk:** Frontend, REST APIs, WebSockets (for both room chats and course help chats), DB constraints, file storage, and analytics must interoperate. Mitigated via modular development, API contracts, socket event specs (course:join, course:message), and iterative testing with Postman and DB query validation.

Conclusion

UIU Dock (UIU Dock) is proposed as a unified, database-driven platform to streamline course learning by converging three essential services — a searchable Resource Center/Hub, course-specific Online Group Study Rooms, and a Help Desk Center implemented as **course-wise group chats** (select a course → enter that course's dedicated chat to discuss problems and solutions) — into one accessible web application. As shown in the Benchmark Analysis, existing platforms support only isolated features, whereas UIU Dock delivers the complete set including study rooms, course help chats, real-time communication, and full database documentation in a single UIU-focused system.

As a Database Management System Lab project, the focus will be on a functional prototype backed by a robust, normalized relational design, well-defined workflows, modular development, real-time but DB-persisted communication per course, and demonstrable analytical SQL queries. The project is practical, scalable for future enhancements, and achievable within the academic timeframe while delivering tangible value for student collaboration and resource management.